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GitHub Link	https://github.com/192425292simats-byte/MLA0403-DEEP-LEARNING

SIMATS ENGINEERING

CO3 - ASSESSMENT TOOL 3

Comparative Analysis

**COURSE NAME:DEEP LEARNING
SLOT :A**

COURSE CODE: MLA0403

COURSE OUTCOME COVERED

CO3: Students will be able to understand, analyze and apply CNN concepts including layers, filters, parameter sharing, regularization, AlexNet and ResNet for real-world image processing applications. (BTL-6)

CO Weightage: 25%

Duration: 90 minutes

Assessment Tool Weightage: 40%

Marks: 25

Code of Conduct:

I (V Sunil Kumar / 192425292) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

Name of the student:V Sunil

Criteria	Selection of Studies/Parameters (2)	Comparison & Differentiation (2.5)	Critical Evaluation (2.5)	Synthesis & Insight Generation (2)	Clarity & Presentation (1)	Total (10)
Weightage	20 %	25%	25%	20 %	10 %	100 %
Q1						
Q2						
Q3						
Q4						
Q5						
Total						

Signature of the Faculty:

Total Marks :

QUESTIONS: 05

Total Marks:25

Q1. CNN vs Fully Connected Neural Network

A medical imaging organization wants to classify X-ray images. A **Convolutional Neural Network (CNN)** is more suitable than a traditional Fully Connected Neural Network (FCNN) because images contain important spatial relationships between neighboring pixels.

Comparison

Parameter	CNN	Fully Connected Neural Network
Architecture	Convolution, activation, pooling and dense layers	Mainly fully connected layers
Input handling	Preserves image structure	Usually converts image into a flat vector
Feature extraction	Automatically learns spatial features	Does not explicitly exploit spatial locality
Parameter count	Relatively low due to parameter sharing	Very high
Spatial information	Preserved through convolution	Lost after flattening
Computational cost	Lower for image feature extraction	Very high for large images
Feature detection	Detects edges, textures and shapes	Learns features without explicit spatial structure
Translation handling	Better tolerance to feature location changes	Poorer
Image suitability	Highly suitable	Less suitable

Role of CNN Components

Convolutional layers:

Apply filters to local image regions and extract visual patterns.

Filters:

Learn patterns such as edges, curves, textures and eventually more complex anatomical structures.

Pooling layers:

Reduce spatial dimensions while retaining important features. This decreases computation and provides some robustness to small changes in object position.

Parameter Difference

Consider a 224×224 grayscale X-ray:

$$224 \times 224 = 50,176 \text{ input pixels}$$

If these pixels are connected directly to 1,000 neurons, approximately:

$$50,176 \times 1,000 = 50,176,000 \text{ weights}$$

would be required, excluding biases.

A 3×3 CNN filter requires only:

$$3 \times 3 = 9 \text{ weights} + \text{bias}$$

and the same filter is reused across the image.

Critical Evaluation

The FCNN can theoretically learn image classifications, but it becomes inefficient as image resolution increases. It also treats pixels more independently and does not naturally exploit local spatial relationships.

CNNs are specifically designed to exploit **local connectivity and parameter sharing**, making them much more efficient for image data.

Conclusion

CNN is clearly preferable for X-ray classification because it preserves spatial information, requires fewer parameters, automatically learns hierarchical features and provides better computational efficiency.

This directly satisfies the question's requested comparison of architecture, motivation, spatial feature extraction, parameters, computational complexity and image handling.

Q2. Different CNN Layers and Filters

For industrial defect detection, different CNN layers perform complementary functions. Early layers identify simple patterns, while deeper layers combine these patterns to recognize complex defects.

Comparison of CNN Layers

Layer	Main Function	Spatial Information	Computation	Role in Defect Detection
Convolution	Feature extraction	Preserves spatial structure	Moderate	Detects edges, textures and defects
Activation	Non-linear transformation	Preserves feature-map structure	Low	Enables complex feature learning
Pooling	Dimension reduction	Reduces spatial resolution	Low	Retains important features and reduces computation
Fully Connected	Classification	Spatial structure mostly summarized	High depending on size	Produces final defect category

Convolutional Layer

The convolution operation applies learnable filters to the image.

For example:

Input image → 3×3 filter → feature map

The filters may detect:

- Edges
- Lines
- Corners
- Surface patterns
- Cracks
- Scratches

Activation Layer

ReLU is commonly used:

$$\text{ReLU}(x) = \max(0, x)$$

It introduces non-linearity and allows the network to learn complicated relationships.

Pooling Layer

A 2×2 max-pooling operation can reduce the feature-map dimensions.

For example:

$$28 \times 28 \rightarrow 14 \times 14$$

This reduces computational requirements while retaining strong features.

Fully Connected Layer

After feature extraction, the learned representation is used for classification, such as:

Normal $\rightarrow 0$

Defective $\rightarrow 1$

or multiple defect classes.

Early vs Deep Filters

Early Filters	Deeper Filters
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Simple features	Complex features
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Edges	Object/defect patterns
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Lines	Shapes
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Corners	Surface structures
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Basic textures	Specific defect combinations
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For example, in a manufacturing system:

Layer 1: detects edges

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Layer 2: detects lines and textures

↓

Layer 3: detects cracks/scratches

↓

Deep layers: recognize complete defect patterns

Critical Evaluation

Using only shallow features may not distinguish between a normal surface and a complex defect. Conversely, excessive depth can increase computation and may increase the risk of overfitting if the dataset is small.

Therefore, the CNN should use sufficient depth to capture defect characteristics while controlling complexity through pooling, batch normalization, regularization and suitable training data.

Conclusion

The layers work together hierarchically: **convolution extracts features, activation enables non-linear learning, pooling reduces dimensionality, and fully connected/output layers perform classification.** This combination improves the ability to identify subtle industrial defects.

The question specifically requires comparison of these layers and early/deep filters, which is addressed above.

Q3. Parameter Sharing vs Independent Weights

A smart surveillance system must process thousands of high-resolution images. Parameter sharing is therefore essential for efficient CNN-based processing.

Comparison

Parameter	CNN Parameter Sharing	FCNN Independent Weights
Weight usage	Same filter reused across image	Different weights for connections
Number of parameters	Much lower	Very high
Memory	Lower	Higher
Computation	More efficient for images	Expensive for high-resolution images
Feature detection	Same feature can be detected anywhere	Location-specific learning
Spatial structure	Exploited	Not naturally exploited
Translation handling	Better	Weaker
Scalability	Better	Poor for large images

Parameter Sharing

Suppose a CNN has a **3×3 filter**.

It contains only:

9 weights + 1 bias

The filter is applied to many different regions:

Region 1 $\rightarrow$ Region 2 $\rightarrow$ Region 3 $\rightarrow$... $\rightarrow$ Region N

The same weights are reused.

Therefore, the network does not need a separate set of weights for every image position.

Independent Weights

In a fully connected network, every input pixel can have a separate connection to every neuron.

For a high-resolution image, this creates a very large number of parameters.

For example, if an image contains:

$1,000 \times 1,000 = 1,000,000$ pixels

and these are connected to 1,000 neurons:

$1,000,000 \times 1,000 = 1$ billion weights

would be required before considering biases.

This creates significant memory and computational requirements.

Translation-Related Feature Detection

Suppose a surveillance camera sees a person's face on the left side of an image and another face on the right side.

A CNN filter trained to detect a particular edge or pattern can recognize it in both locations because the same filter is reused.

Thus:

Same feature + different position → same filter can detect it

This gives CNNs useful translation-related robustness.

Critical Evaluation

Parameter sharing dramatically improves learning efficiency, but it does not make CNNs completely invariant to every transformation. Large rotations, scale changes, viewpoint changes and severe illumination changes may still require augmentation or more advanced architectures.

Conclusion

Parameter sharing is extremely important for large-scale surveillance because it:

- Reduces parameters.
- Reduces memory.
- Reduces computation.
- Reuses learned visual features.
- Improves feature detection across image locations.
- Makes high-resolution image processing more practical.

Therefore, CNN parameter sharing is much more appropriate than independent fully connected weights for thousands of high-resolution images.

This addresses all comparison parameters specified in the question.

Q4. AlexNet vs ResNet

A computer vision company is choosing between AlexNet and ResNet for a complex image classification system.

Comparison

Parameter	AlexNet	ResNet
Architecture	Traditional CNN	Residual CNN
Depth	Relatively shallow	Can be very deep
Layer organization	Sequential convolution/pooling/dense structure	Residual blocks with shortcut connections
Filters	Learn hierarchical visual features	Learn deeper hierarchical features
Parameters	Large, especially dense layers	Varies by version; generally more efficient for deep feature learning
Feature extraction	Good	Very strong
Training	Relatively straightforward	Deep models made easier to train using residual connections
Vanishing gradient	More manageable because network is shallower	Skip connections greatly reduce degradation/gradient problems
Computational cost	Lower than many deep ResNet models	Depends on ResNet version
Classification performance	Strong historical baseline	Generally stronger on complex recognition tasks

AlexNet

AlexNet was an important CNN architecture that demonstrated the effectiveness of deep convolutional learning for image recognition.

Its architecture includes convolutional layers followed by deeper feature-processing and classification layers.

Advantages:

- Relatively simple architecture.
- Good historical baseline.
- Lower complexity than many modern deep networks.

Limitations:

- Less depth.
- Less powerful hierarchical feature representation compared with modern architectures.
- Large fully connected layers can require many parameters.
- Not designed specifically to solve the optimization challenges of very deep networks.

ResNet

ResNet introduces **residual/shortcut connections**.

Instead of forcing every layer to learn an entirely new transformation, a residual block learns a modification to the input:

$$\text{Output} = F(x) + x$$

where:

- x = input
- $F(x)$ = learned residual function

This makes deep networks easier to optimize.

Vanishing-Gradient Problem

As networks become deeper, gradients can become extremely small during backpropagation.

This can make learning difficult.

ResNet addresses this through shortcut connections, allowing information and gradients to travel more directly through the network.

Critical Evaluation

AlexNet is useful as a baseline and may require fewer computational resources. However, for **complex image recognition**, the deeper feature-learning capability of ResNet provides a major advantage.

The choice should still consider hardware constraints. A very deep ResNet may be unnecessarily expensive for a simple classification task.

Recommendation

ResNet is the more suitable architecture for complex image recognition.

Reasons:

1. Better deep feature extraction.
2. Residual connections improve training.
3. Reduced degradation/vanishing-gradient issues.
4. Better scalability to deeper architectures.
5. Strong classification capability.

A suitable ResNet variant should be selected based on the available computational resources and required inference speed.

The comparison directly follows the parameters specified in the assessment: depth, organization, filters, parameters, feature extraction, training, gradient problems, cost and performance.

Q5. Regularized CNN vs Non-Regularized CNN

An agricultural organization has developed a CNN for plant disease classification. The model has **very high training accuracy but poor performance on unseen images**.

This is a classic indication of **overfitting**.

Comparison

Parameter	Non-Regularized CNN	Regularized CNN
Training accuracy	Often very high	High but more controlled
Test accuracy	May be poor	Usually better
Overfitting	High risk	Reduced
Generalization	Poor	Better
Training stability	Can be unstable	Improved with suitable normalization
Data requirement	More likely to memorize training data	Better use of varied training examples
Computational requirement	Lower initially	Slightly higher due to augmentation/dropout
Real-world performance	Less reliable	More reliable

1. Dropout

Dropout randomly disables neurons during training.

For example:

Dense Layer → Dropout(0.5) → Output

This prevents the network from depending too heavily on particular neurons.

Benefit: Reduces overfitting.

Limitation: Excessive dropout can make learning slower or cause underfitting.

2. Data Augmentation

Leaf images can be modified using appropriate transformations such as:

- Rotation
- Scaling
- Cropping
- Brightness changes
- Contrast changes
- Flipping where biologically appropriate

For example:

Original leaf image

- rotation
- brightness variation
- scale variation
- additional training example

This teaches the CNN to recognize disease patterns rather than memorize individual images.

3. Batch Normalization

Batch normalization normalizes intermediate activations during training.

It can:

- Stabilize training.
- Help faster convergence.
- Make optimization easier.
- Provide some regularization effect.

Critical Evaluation

The major problem is not that the CNN is incapable of learning; rather, it is **memorizing the training dataset**.

Therefore, simply increasing the network size is not the correct solution.

The organization should first improve the training strategy and dataset diversity.

Recommended Approach

A combined strategy is most appropriate:

Data Augmentation + Batch Normalization + Dropout

Data Augmentation

Should be strongly emphasized because real-world leaf images can vary in:

- Orientation
- Lighting
- Scale
- Background
- Image quality

Batch Normalization

Should be included within the CNN architecture to improve training stability.

Dropout

Can be applied before the final classification layer to reduce over-dependence on specific learned features.

Additional techniques such as **early stopping and weight decay** can also be considered if overfitting continues.

Example Architecture

Input Leaf Image

↓

Convolution

↓

Batch Normalization

↓

ReLU

↓

Pooling

↓

Convolution

↓

Batch Normalization

↓

ReLU

↓

Pooling

↓

Flatten / Global Average Pooling

↓

Dropout

↓

Fully Connected

↓

Disease Classification

Final Recommendation

A **regularized CNN** is clearly more appropriate for reliable plant-disease classification.

The most important combination is:

Data Augmentation → Batch Normalization → Dropout

This improves generalization and makes the model more robust to unseen leaf images.

The question specifically asks for comparison based on overfitting, generalization, training stability, computation and performance, followed by a justified recommendation.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Selection of Studies/Parameters	20%	Selects highly relevant and diverse studies with well-defined comparison parameters	Selects relevant studies with minor gaps in parameter definition	Limited or partially relevant selection	Poor or inappropriate selection of studies
Comparison & Differentiation	25%	Clearly compares multiple aspects (methods, results, limitations) with strong differentiation	Good comparison with minor gaps	Basic comparison with limited depth	No clear comparison; mostly descriptive
Critical Evaluation	25%	Critically evaluates strengths, weaknesses, and implications with strong justification	Provides evaluation with some justification	Limited evaluation; lacks depth	No critical evaluation provided
Synthesis & Insight Generation	20%	Generates meaningful insights and conclusions from comparisons	Some insights derived from comparison	Limited or unclear insights	No insights generated
Clarity	10%	Information is clearly presented (tables/figures) with logical structure	Generally clear with minor issues	Presentation lacks clarity or structure	Poor presentation and difficult to understand